

Vol 7 Chapter 4 — Magnetostatics

Keeper (author pass — deep math/physics revision)

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Chapter 4 — Magnetostatics

Level 1 — one sentence

Magnetic fields are the substrate's $U(1)$ gauge connection sourced by *moving* $SO(2)$ -charges (currents) — the time-independent limit of Maxwell's equations with steady currents $\vec{J}$ gives $\nabla \times \vec{B} = \mu_0 \vec{J}$ (Ampère) and $\nabla \cdot \vec{B} = 0$ (no monopoles), with the magnetic field $\vec{B} = \nabla \times \vec{A}$ derivable from the vector potential $\vec{A}$ (the substrate's spatial $U(1)$ connection components).

Level 2 — graduate-physicist precision

4.1 The magnetostatic equations

For steady currents (time-independent ρ and $\vec{J}$ with $\nabla \cdot \vec{J} = 0$):

$$\nabla \cdot \vec{B} = 0 \quad (\text{no magnetic monopoles})$$

$$\nabla \times \vec{B} = \mu_0 \vec{J} \quad (\text{Ampère's law})$$

(displacement current $\mu_0 \epsilon_0 \partial_t \vec{E} = 0$ in the static case).

The first equation $\nabla \cdot \vec{B} = 0$ implies $\vec{B}$ is the curl of some vector field:

$$\vec{B} = \nabla \times \vec{A}$$

where $\vec{A}$ is the vector potential (the spatial part of the 4-potential $A^\mu = (\phi/c, \vec{A})$).

4.2 Biot-Savart law

For a current-carrying element $I d\vec{\ell}'$ at position $\vec{r}'$:

$$d\vec{B}(\vec{r}) = \frac{\mu_0 I}{4\pi} \frac{d\vec{\ell}' \times (\vec{r} - \vec{r}')}{|\vec{r} - \vec{r}'|^3}$$

Integrating over the current distribution gives the total field. Biot-Savart is to magnetostatics what Coulomb's law is to electrostatics — the explicit source-to-field formula.

For a long straight wire: $B(r) = \mu_0 I / (2\pi r)$. For a circular loop of radius R at distance z on its axis: $B(z) = \mu_0 I R^2 / [2(R^2 + z^2)^{3/2}]$.

Substrate-mechanism reading: a current is a substrate K-type configuration of moving $SO(2)$ -charged K-types; their motion induces an extended $SO(2)$ -connection field in space — the magnetic field.

4.3 Ampère's law applications

Integral form: $\oint \vec{B} \cdot d\vec{\ell} = \mu_0 I_{\text{enc}}$.

Useful for symmetric current distributions: - Infinite straight wire: $B = \mu_0 I / (2\pi r)$ - Solenoid (long, n turns/m): $B = \mu_0 n I$ inside, $B = 0$ outside - Toroid: $B = \mu_0 N I / (2\pi r)$ inside the toroid

4.4 Magnetic vector potential

From $\vec{B} = \nabla \times \vec{A}$ and Ampère + $\nabla \cdot \vec{B} = 0$:

$$\nabla \times \vec{B} = \nabla \times (\nabla \times \vec{A}) = \nabla(\nabla \cdot \vec{A}) - \nabla^2 \vec{A} = \mu_0 \vec{J}$$

In Coulomb gauge ($\nabla \cdot \vec{A} = 0$):

$$\nabla^2 \vec{A} = -\mu_0 \vec{J}$$

(component-wise Poisson equation). Solution:

$$\vec{A}(\vec{r}) = \frac{\mu_0}{4\pi} \int \frac{\vec{J}(\vec{r}')}{|\vec{r} - \vec{r}'|} d^3 r'$$

The vector-potential form of the magnetic field.

4.5 Magnetic dipole

A small current loop (area A , current I) has magnetic dipole moment $\vec{m} = IA\hat{n}$. Vector potential at distance r :

$$\vec{A}_{\text{dipole}}(\vec{r}) = \frac{\mu_0}{4\pi} \frac{\vec{m} \times \vec{r}}{r^3}$$

Magnetic field:

$$\vec{B}_{\text{dipole}}(\vec{r}) = \frac{\mu_0}{4\pi} \frac{3(\vec{m} \cdot \hat{r})\hat{r} - \vec{m}}{r^3}$$

Decreases as $1/r^3$ (vs $1/r^2$ for point charge). Spinning electrons have intrinsic magnetic dipole moments via their substrate Pin(2) coupling (Volume 5 Chapter 3 spin).

4.6 Magnetic monopoles

Maxwell's equations as written have $\nabla \cdot \vec{B} = 0$ — no magnetic monopoles. Dirac 1931 showed that if even one magnetic monopole existed, the electric charge would be quantized:

$$eg = nh/2$$

with g the magnetic charge, n integer. Since electric charges *are* quantized, this is suggestive.

But: no magnetic monopole has ever been observed. BST predicts no monopoles as part of the Five-Absence Predictions Set (Casey-named principle #2, Volume 0 Chapter 3). The substrate's $U(1)$ structure forbids monopoles structurally: only $SO(2)$ -charged K-types exist, and the substrate's K-type topology doesn't admit pure-magnetic-source configurations.

If a magnetic monopole is ever detected, BST is falsified.

4.7 Force on currents (Lorentz)

A charge q moving with velocity $\vec{v}$ in field $\vec{B}$:

$$\vec{F} = q\vec{v} \times \vec{B} \quad (\text{magnetic Lorentz})$$

Combined with electric: $\vec{F} = q(\vec{E} + \vec{v} \times \vec{B})$. This is the Lorentz force law.

For a current-carrying wire of length ℓ with current I in field $\vec{B}$: $\vec{F} = I\vec{\ell} \times \vec{B}$.

Substrate reading: the Lorentz force is the substrate's $SO(2)$ -charge K-type coupling to the substrate $U(1)$ field; for moving charges, the magnetic ($\vec{v} \times \vec{B}$) component is the substrate's relativistic mixing of electric and magnetic field components (Chapter 7).

4.8 Worked example: Earth's magnetic field

Earth's surface magnetic field $\sim 25 - 65 \mu\text{T}$. Approximate as magnetic dipole with $m \approx 8 \times 10^{22} \text{ A}\cdot\text{m}^2$. At distance r from center (radius $R_E = 6.4 \times 10^6 \text{ m}$), equatorial field:

$$B(R_E) = \frac{\mu_0 m}{4\pi R_E^3} = \frac{(4\pi \times 10^{-7})(8 \times 10^{22})}{4\pi(6.4 \times 10^6)^3} \approx 3 \times 10^{-5} \text{ T} = 30 \mu\text{T}$$

Matches observation (Earth's field is more complex — non-pure-dipole — but the dipole approximation is good to 10-20%).

The dipole moment originates from Earth's liquid-iron outer core convection (dynamo). Substrate reading: the dynamo is a substrate K-type collective magnetic-cluster amplification in the convecting iron K-type configuration.

4.9 K-audit anchors

- Chapter 2: Maxwell's equations from substrate $U(1)$
- Five-Absence Predictions Set (Casey-named #2, Volume 0): no monopoles
- T2470 (Lyra Friday May 22, 2026): substrate $SO(2)$ charge structure

Level 3 — 5th-grader accessibility

Static magnetic fields come from moving charges — currents. A wire carrying current I has a magnetic field that wraps around it at distance r : $B = \mu_0 I / (2\pi r)$. Ampère's law: integrate B around a loop, you get μ_0 times the enclosed current. Useful for solenoids ($B = \mu_0 nI$ inside, zero outside), toroids, etc. Magnets are tiny current loops at the atomic level (spinning electrons). No magnetic monopoles: you can't find a "north pole" without a "south pole" attached. BST predicts no monopoles ever exist (one of the Five-Absence Predictions). If anyone ever finds one, BST is wrong.

What comes next

Chapter 5 develops electromagnetic waves — the propagating solutions to the time-dependent Maxwell equations.

Where to look this up

- Magnetostatics: Griffiths Ch 5; Jackson Ch 5
- Dirac monopole: Dirac 1931 *Proc Roy Soc*
- BST anchors: Five-Absence Predictions; Chapter 2 (Maxwell from substrate)